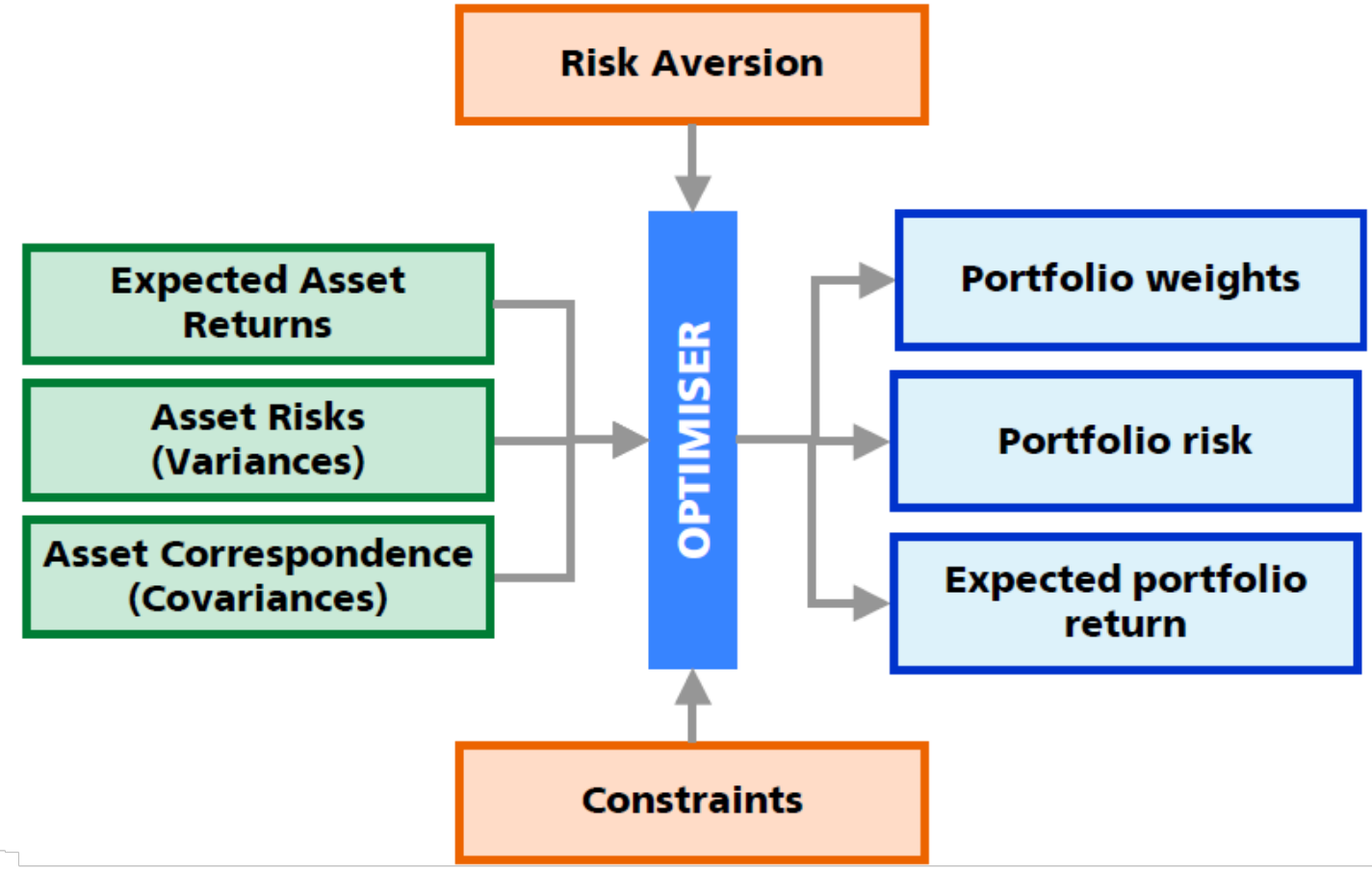
Mean-Variance Optimization and Enhancement

What is Portfolio Optimization



Minimum Variance Portfolio

- Investment universe consists of N risky assets
- The portfolio with the lowest variance is called the Minimum Variance portfolio
- For the unconstrained minimum variance portfolio each asset must have identical marginal contribution to risk

$$\boxed{\mathbf{w}_{\text{Min}} = \frac{\mathbf{V}^{-1} \cdot \mathbf{1}}{\mathbf{1}' \cdot \mathbf{V}^{-1} \cdot \mathbf{1}}}$$

Minimum Variance Portfolio - Example

Minimum Variance Portfolio

	Unconstraint Weights	MCTR
A	-0.89%	7.696%
B	3.60%	7.696%
C	0.72%	7.696%
D	19.60%	7.696%
E	76.96%	7.696%
sum	100.0%	
Variance	0.59%	
Volatility	7.696%	
Return	4.36%	

	Return	Risk
A	11%	25%
B	9%	24%
C	10%	18%
D	5%	11%
E	4%	8%

Correlation Matrix

	A	B	C	D	E
A	1.0	0.5	0.5	0.4	0.2
B	0.5	1.0	0.4	0.2	0.2
C	0.5	0.4	1.0	0.5	0.3
D	0.4	0.2	0.5	1.0	0.5
E	0.2	0.2	0.3	0.5	1.0

Maximize Sharpe Ratio Portfolio

- Sharpe ratio measures the risk adjusted return for asset/portfolio. Generally, the higher the Sharpe ratio, the more attractive the risk-adjusted return.

- **Expected return**: $\mathbf{w}^T\boldsymbol{\mu}$
- **Volatility**: $\sqrt{\mathbf{w}^T\boldsymbol{\Sigma}\mathbf{w}}$
- **Sharpe Ratio (SR)**: expected return per unit of risk

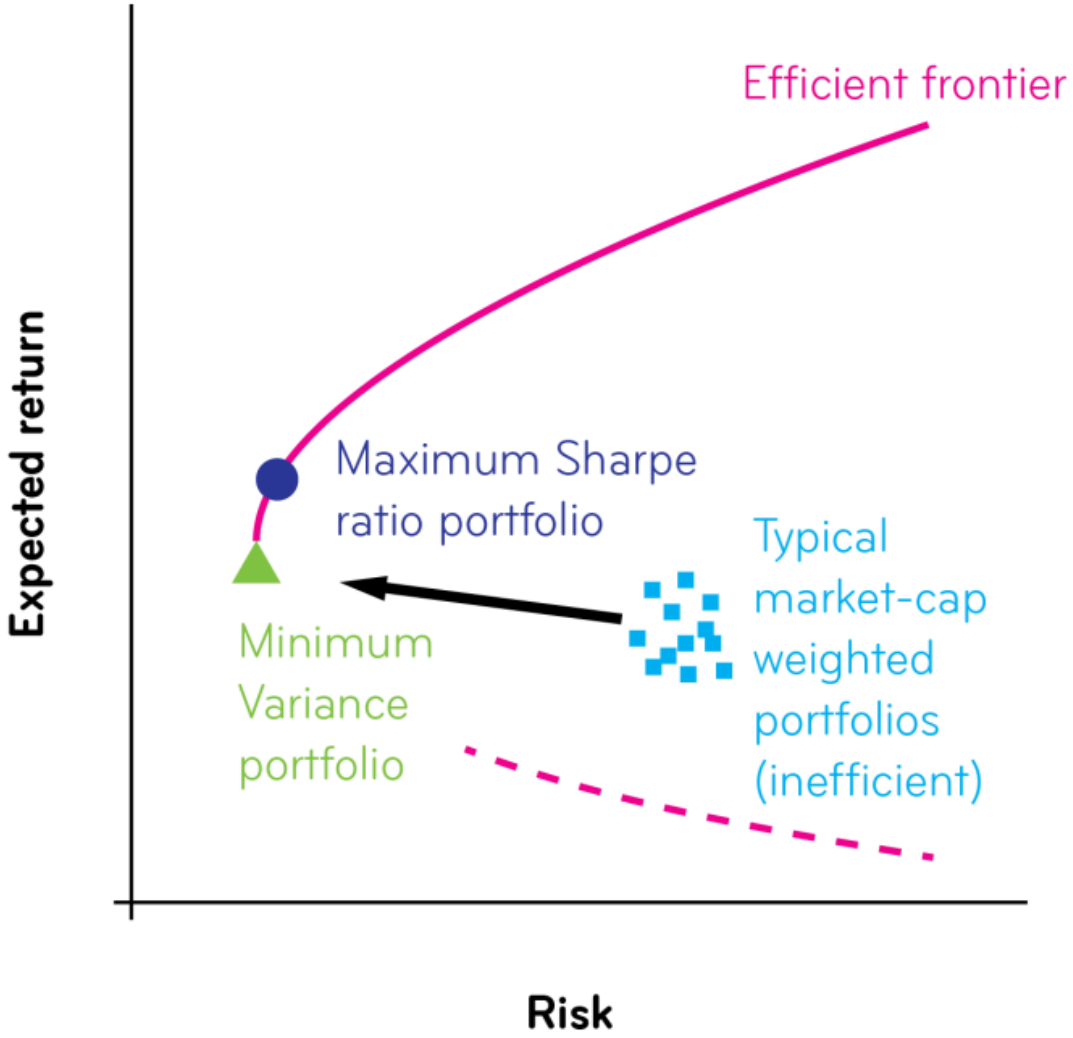
$$\mathrm{SR} = \frac{\mathbf{w}^T\boldsymbol{\mu} - r_f}{\sqrt{\mathbf{w}^T\boldsymbol{\Sigma}\mathbf{w}}}$$

where r_f is the risk-free rate (e.g., interest rate on a three-month U.S. Treasury bill).

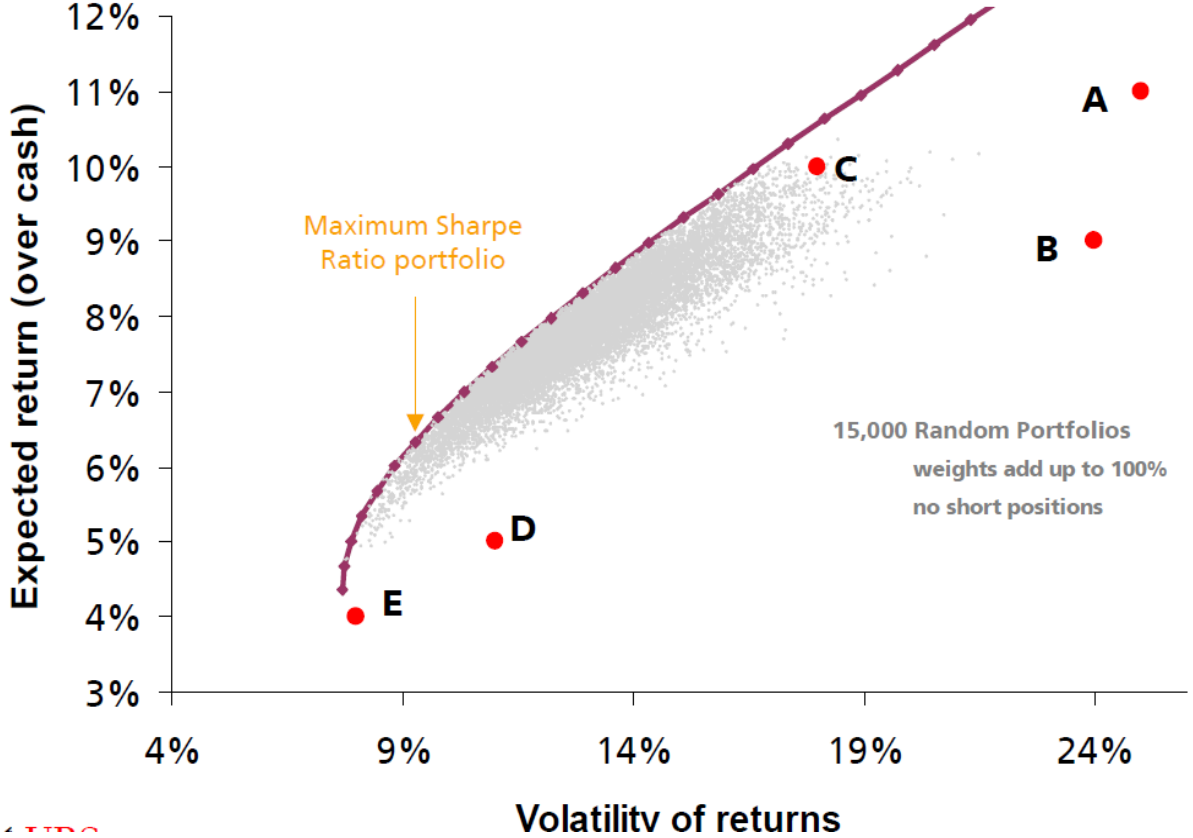
Efficient Frontier

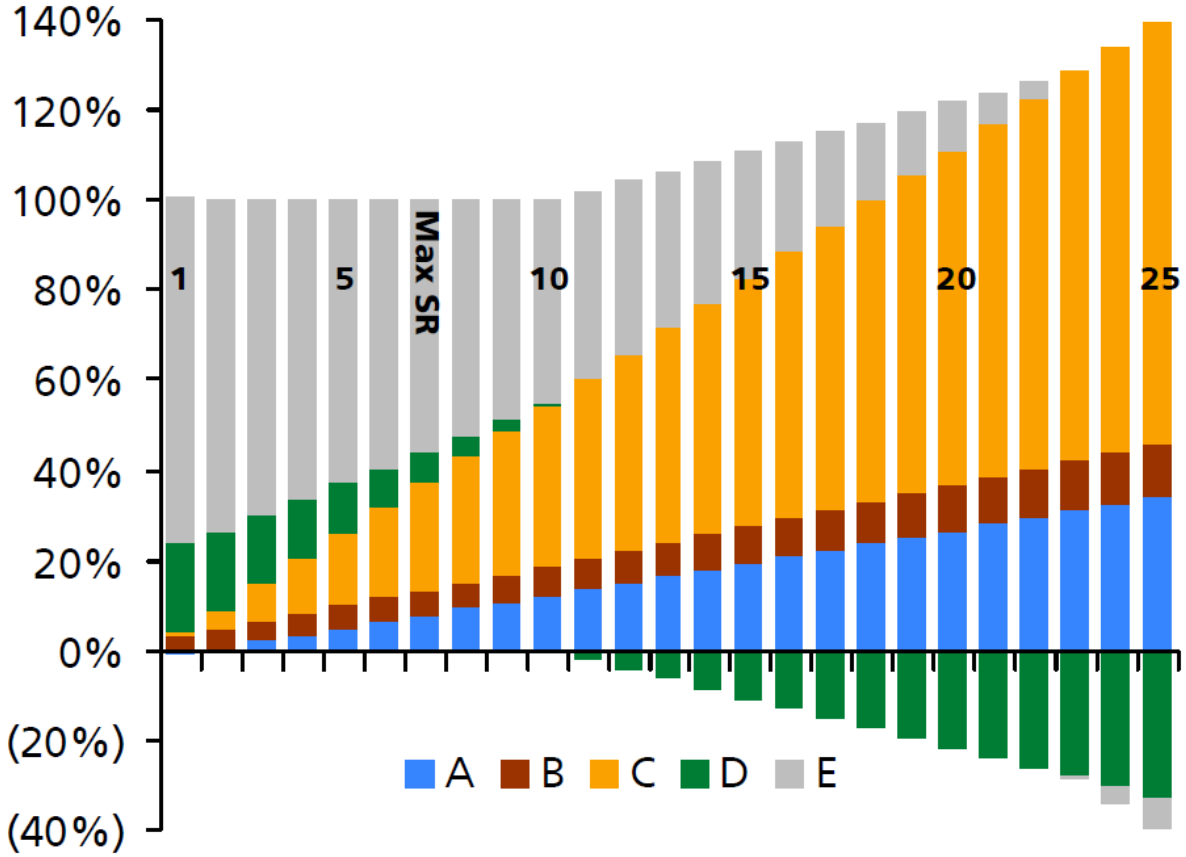
- Portfolios on the efficient frontier are efficient, in the sense that they have the best possible expected return for their level of risk. Such portfolios (without including the risk-free asset) can be plotted in risk (i.e., volatility) – expected return space.

- The efficient frontier together with the minimum variance locus (i.e., the bottom portion of the hyperbola) form the 'upper border' and 'lower border' lines of the feasible set. To the right the feasible set is determined by the envelope of all pairwise asset frontiers.

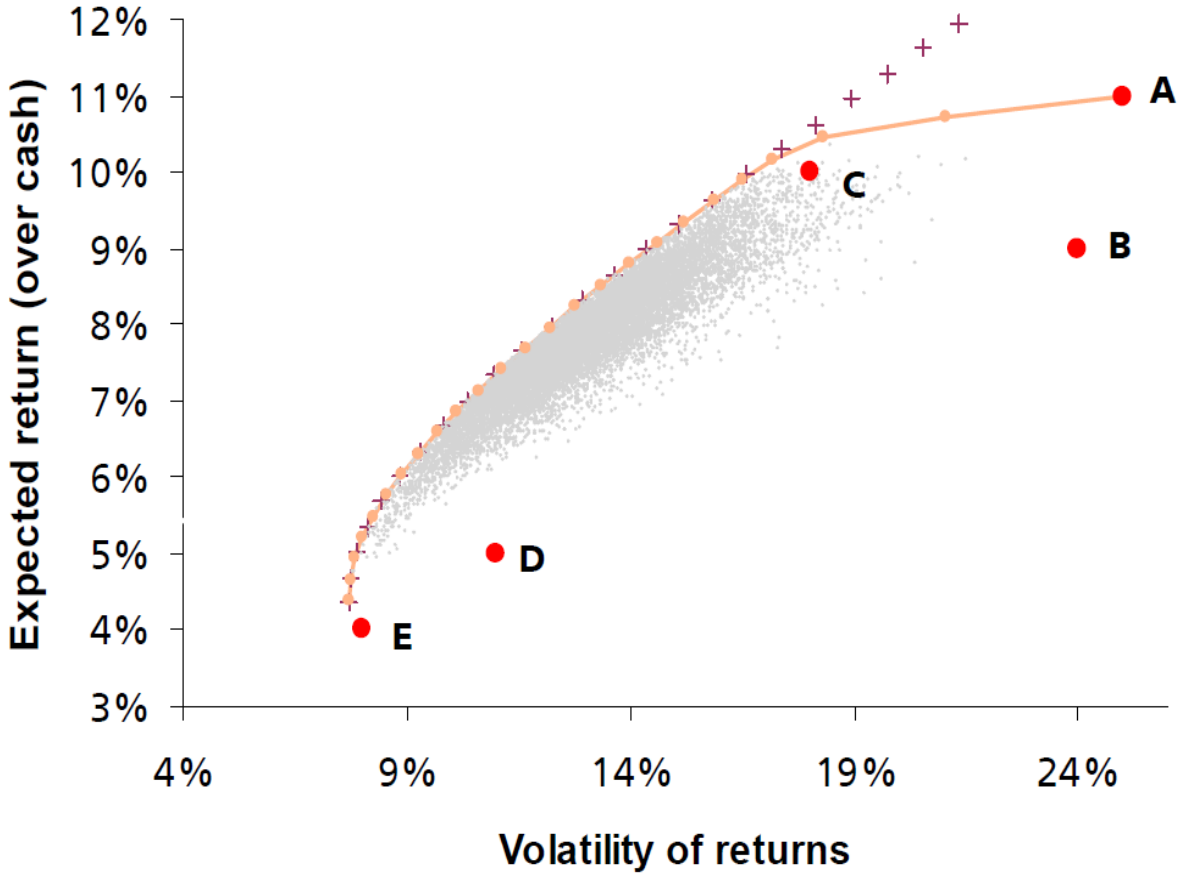


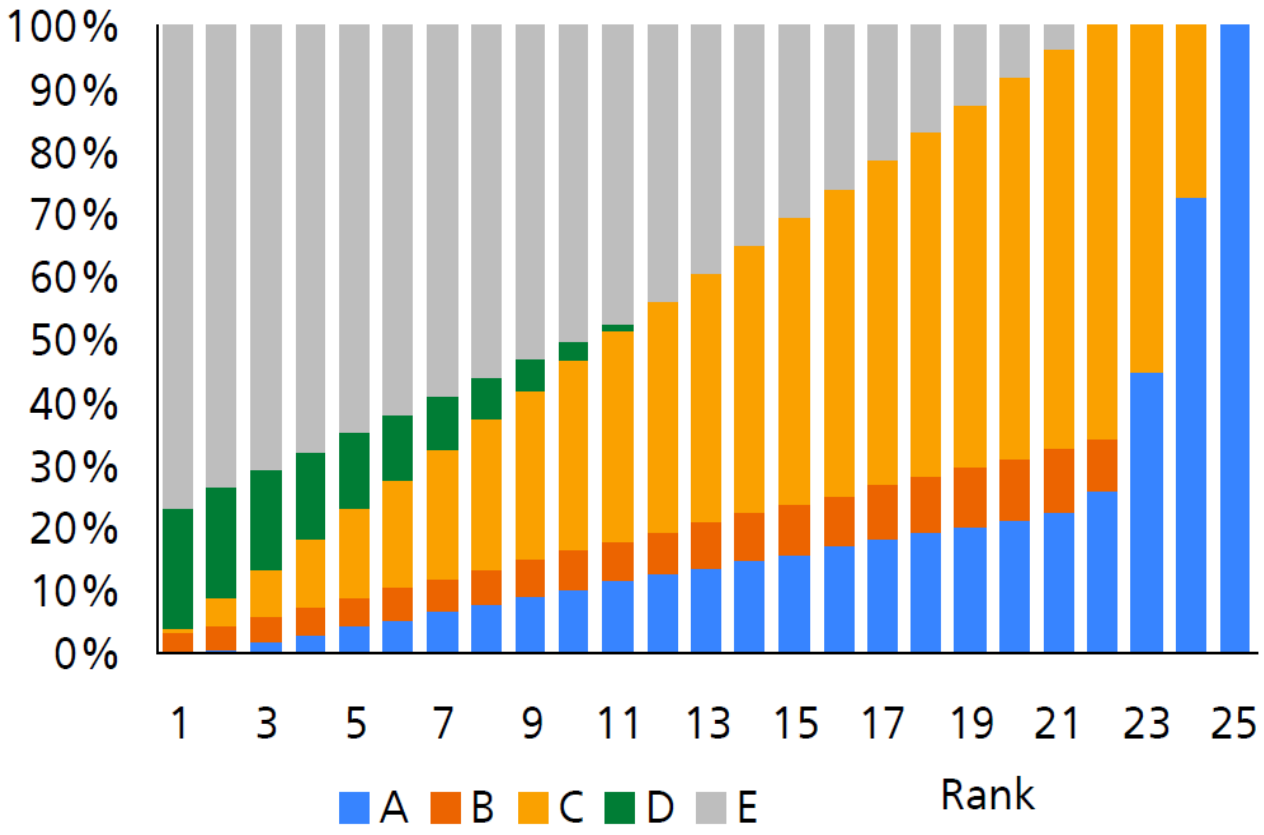
Efficient Frontier Example (unconstrained)





Efficient Frontier Example (constrained)





Utility Function – Cost of Risk

- Portfolio construction is all about trading risk against return
- In the Markowitz world the investor selects a portfolio with minimum risk (volatility) for a given expected return
- In the Markowitz world the cost of risk is proportional to variance
- The exact cost depends on how risk averse you are, introducing λ as a measure for risk tolerance

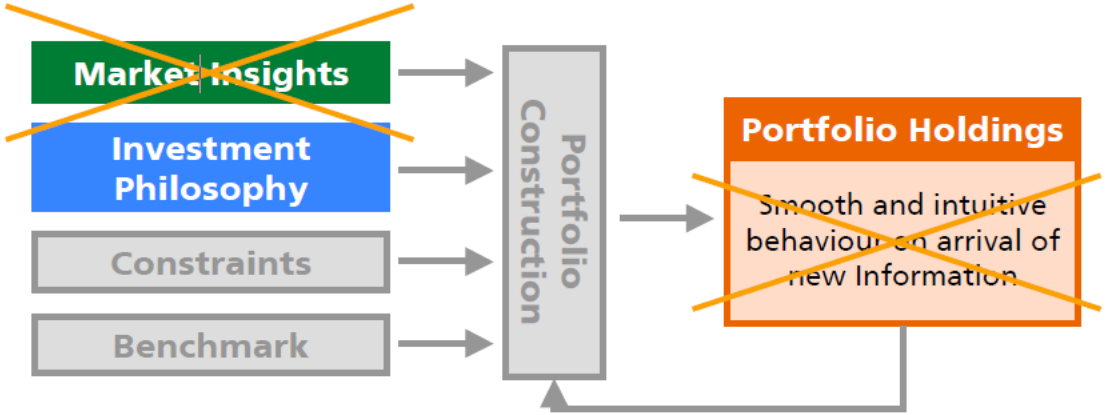
$$\text{Cost of Risk} = \frac{1}{2\lambda}\sigma_P^2$$

- Adjusted for the subjective cost of risk the portfolio return reduces to

$$\boxed{\text{utility} = \mu_p - \frac{1}{2\lambda}\sigma_P^2}$$

MVO efficiency in practice

- Mean variance efficient portfolios are often extreme, unintuitive and input sensitive
- Updating on arrival of new information results in huge shifts in portfolio weights and therefore high costs
- Standard MV analysis forces the user to enter an exact forecast for the return on all assets but most users will have views on some assets and are neutral on others Market



What are the sources of risk?

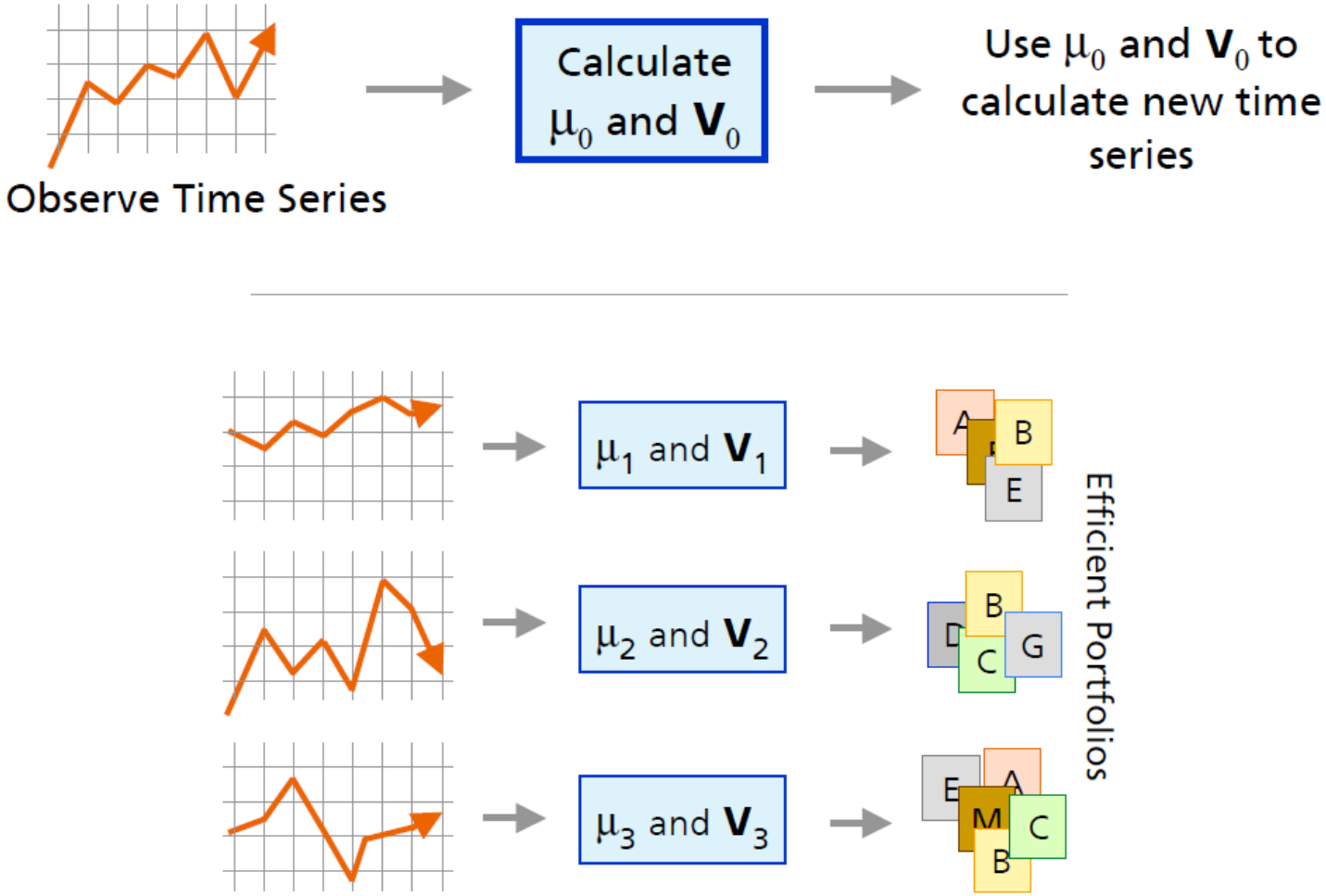
- MV efficient portfolio face two different types of risk
- So far we have assumed that we know μ and V with certainty
- How accurate does μ describe the mean of future asset returns?
- How accurate does V describe the future relation between assets?

Forecasted Uncertainty	Forecast Error
Described by **V**	♦ How accurate is μ ? ♦ How accurate is **V** ?

A way to deal with estimation errors for MVO

- Modify the optimizers inputs slightly and then interpret the outputs
- Resampling
  - Presupposes knowledge of distribution (assume normally distributed returns in most cases)
  - Estimate mean, variance and covariance of a set of historic returns
  - Create returns series by repeatedly drawing from input distribution described above
- Bootstrapping
  - Repeatedly sampling from historical distribution
  - Start with monthly returns
  - Sample 12 monthly returns to create annual return
  - Repeat this over and over to generate distribution

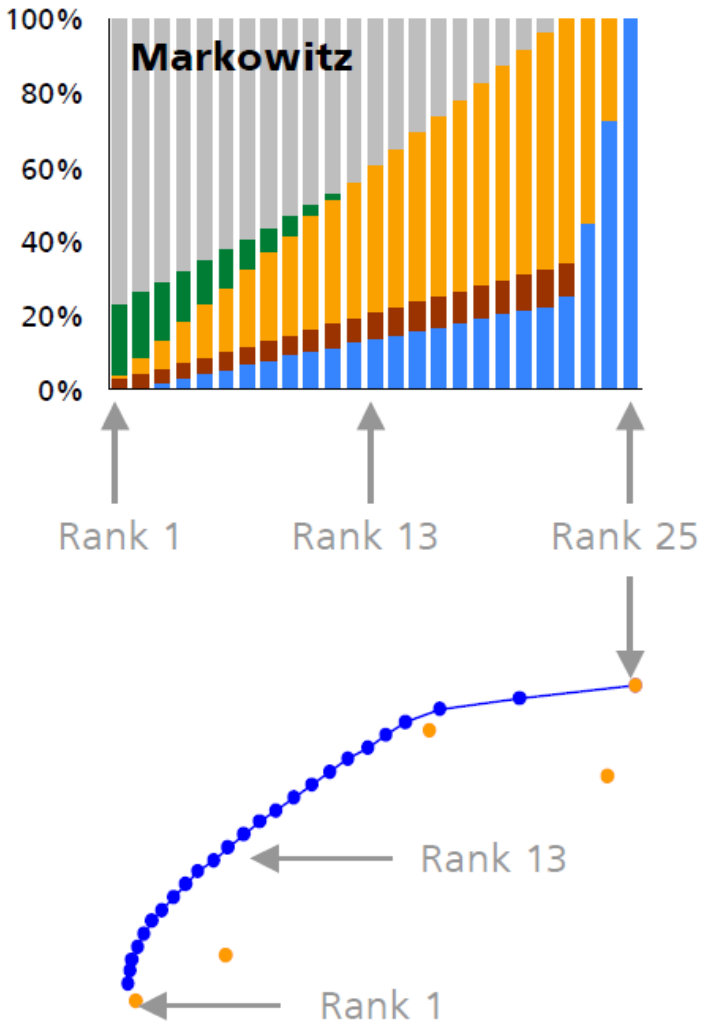
Resampling

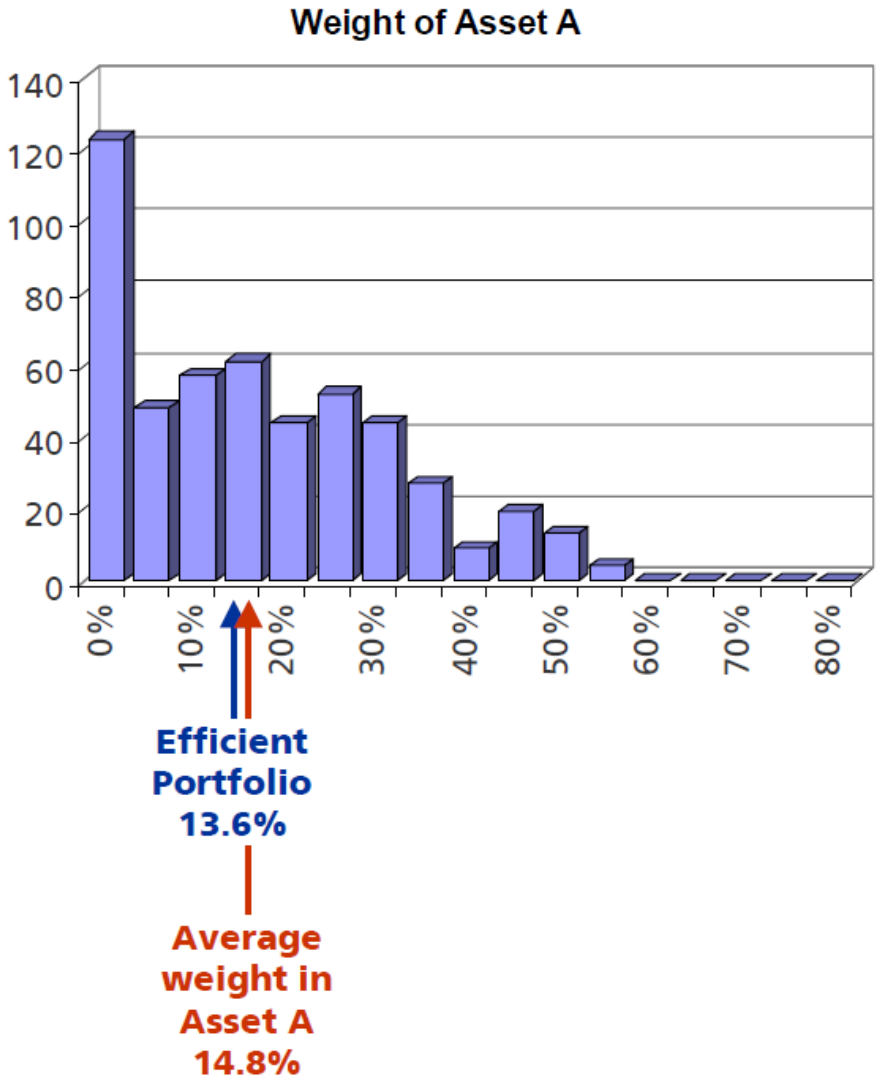

Observe Time Series
Calculate μ_0 and $\mathbf{V}_0$
Use μ_0 and $\mathbf{V}_0$ to calculate new time series
μ_1 and $\mathbf{V}_1$
μ_2 and $\mathbf{V}_2$
μ_3 and $\mathbf{V}_3$
A
B
E
B
D
C
G
E
A
M
C
B
Efficient Portfolios

Resampling – example

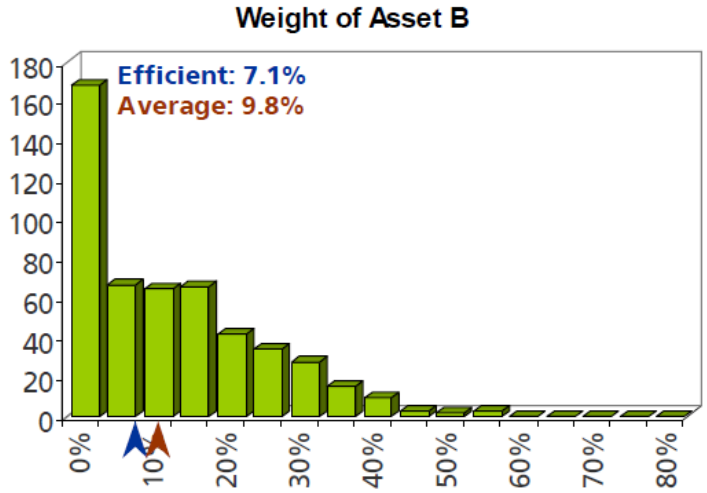
- Assume that μ_0 and V_0 were calculated using 5 years of monthly returns
- We resample by drawing 500 times from the distribution given by μ_0 and V_0
- For each new distribution (μ_i, V_i) we calculate 25 efficient portfolios (from minimum variance portfolio to maximum mean portfolios)

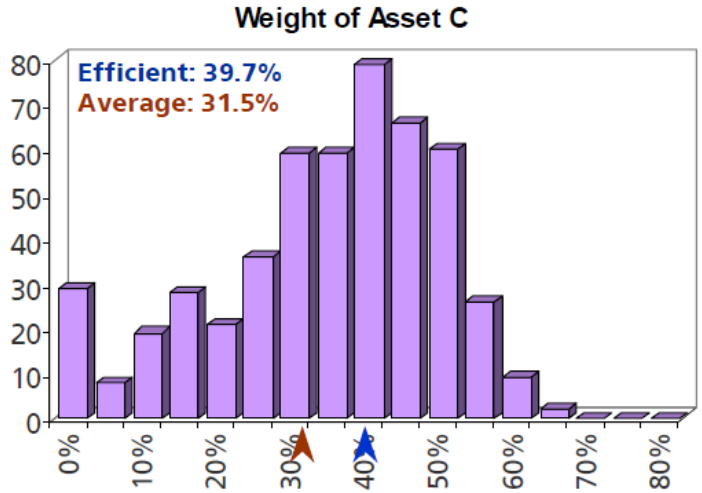
Distribution of Portfolio Weights

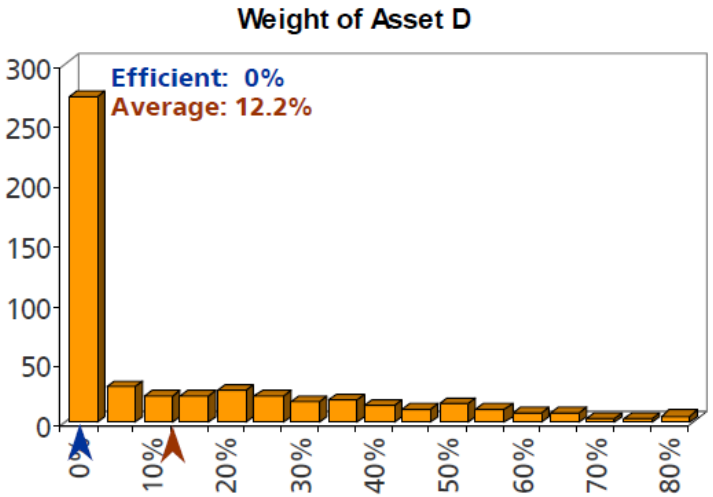


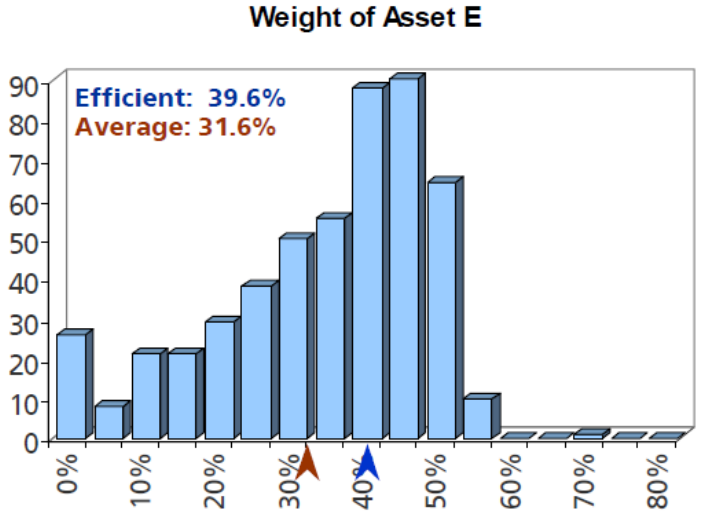


Distribution of Portfolio Weights









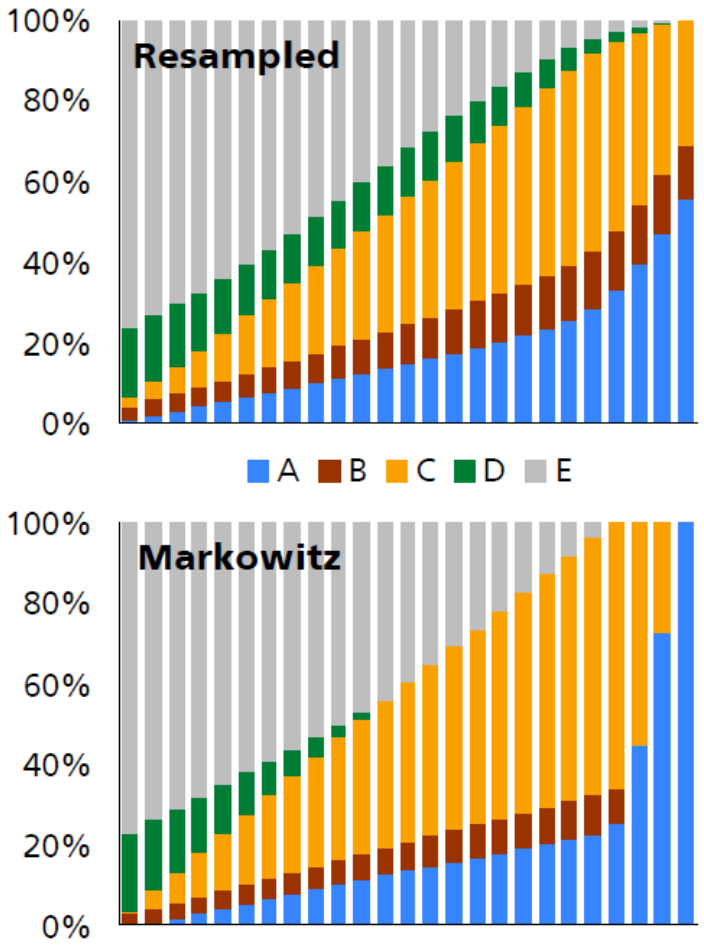
Which portfolios are not efficient?

- When is there enough evidence to conclude that our portfolio is not efficient?
- From the previous distributions we can conclude that for 80% of the 501 resampled portfolios the portfolio weight of
  - Asset A lies between 0% and 34.1%
  - Asset B lies between 0% and 26.1%
  - Asset C lies between 9.3% and 48.9%
  - Asset D lies between 0% and 45.0%
  - Asset E lies between 8.9% and 46.6%
- If we decide that there is enough statistical evidence for a portfolio to be non-efficient if one of the weights is outside the above interval then 239 of the 501 portfolios are non-efficient
- This approach is intuitive but quite heuristic

Resampled Frontier

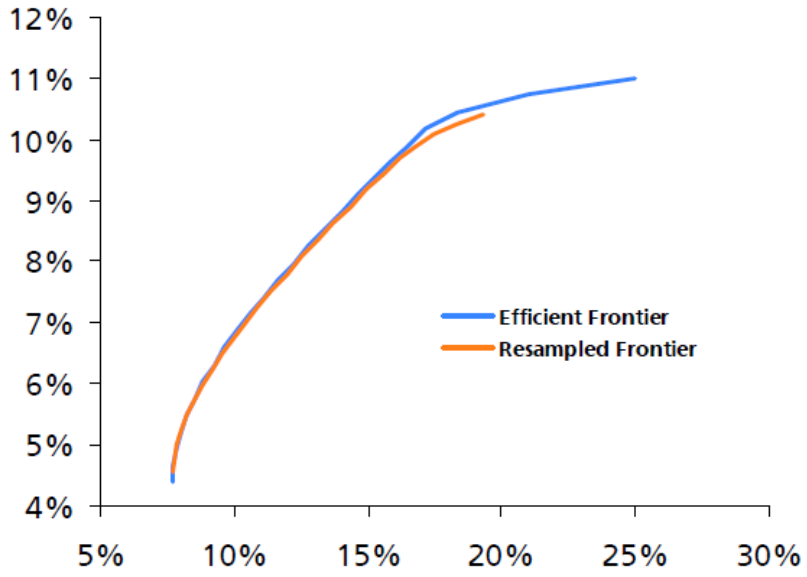
- Resampling leads to distribution of portfolio weights
- We can average the portfolio weights of rank equivalent portfolios
- These resampled portfolios form the Resampled Efficient Frontier
- In order to compare the qualities of the many portfolios, we need to know the structure of the market
- Assume that the original inputs ($\mu 0$, $V0$) describe the market accurately
- We then calculate the expected return and risk of each portfolio

Resample Frontier

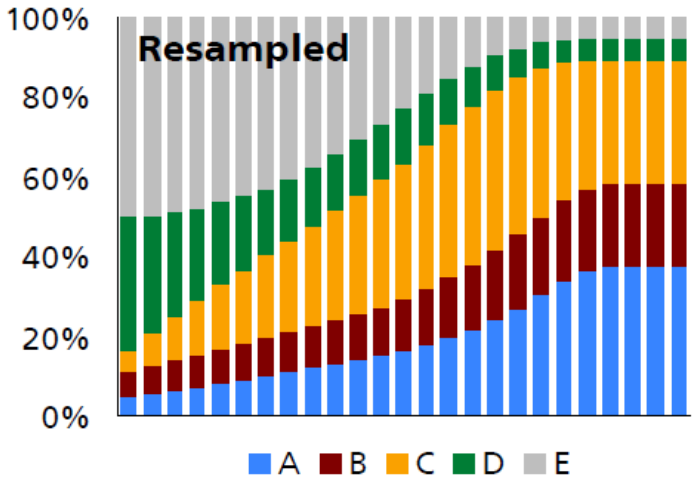


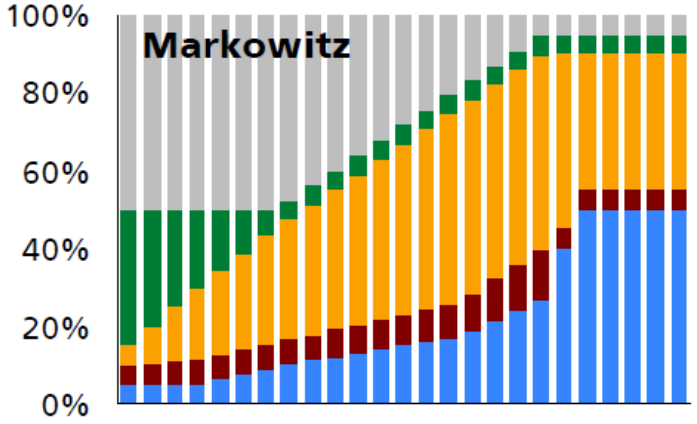
Constraints

- weights add up to 100%
- no short positions



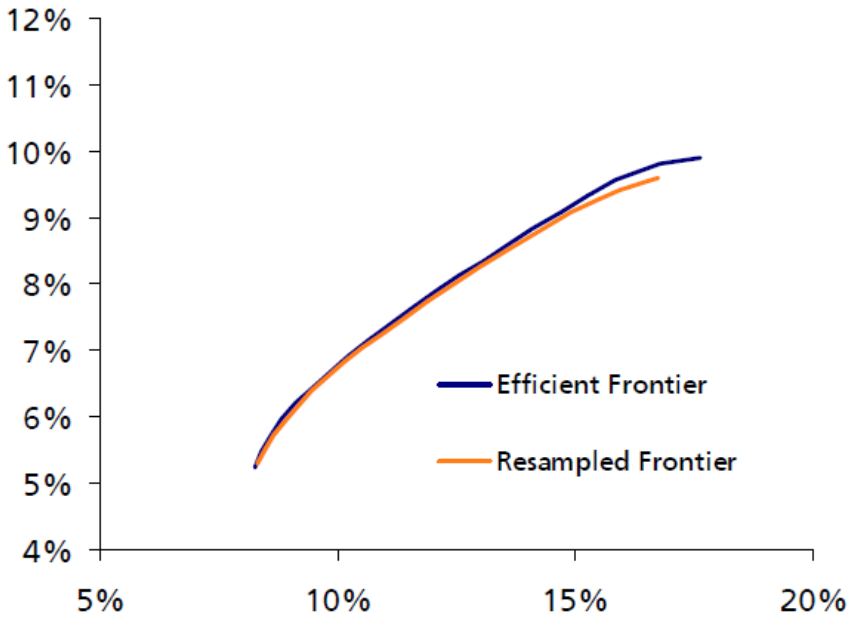
Resample Frontier





Constraints

- weights add up to 100%
- 5% minimum weight in each share
- 50% maximum weight in each share



Resampled Portfolio - conclusion

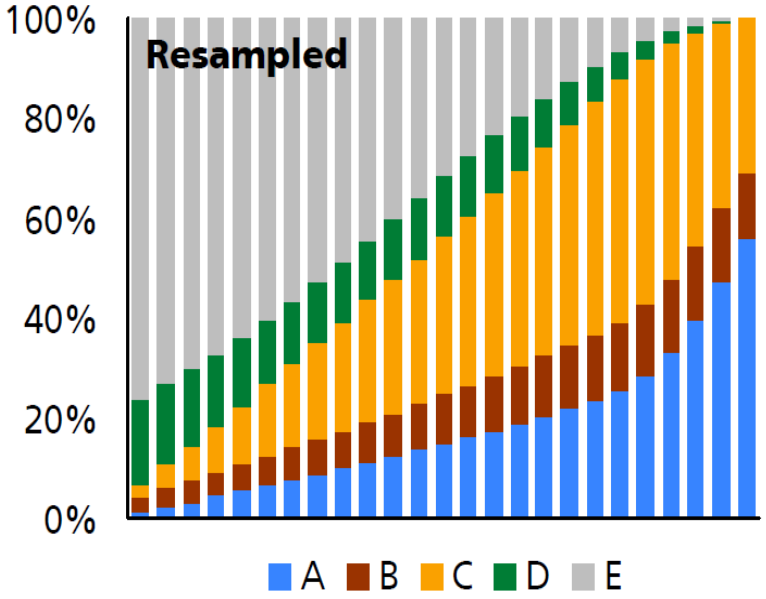
- Resampled and mean-variance efficient portfolios are similar in risk-return space
- Resampled portfolios differ from MV efficient portfolios in weight space (Resampled portfolios more diversified)
- Resampling produces smoother change in portfolio weights if return requirement changes (λ factor in utility function)
- Resampling can be used to decide whether portfolios are statistically different
- Efficient portfolios are close to resampled portfolios
- Resampled portfolio have good statistical properties (mean of portfolios)

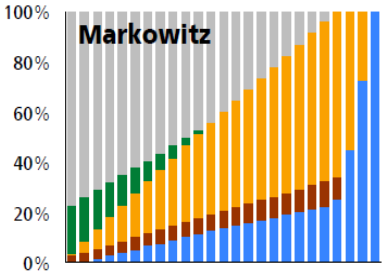
Resampled Portfolio - shortcomings

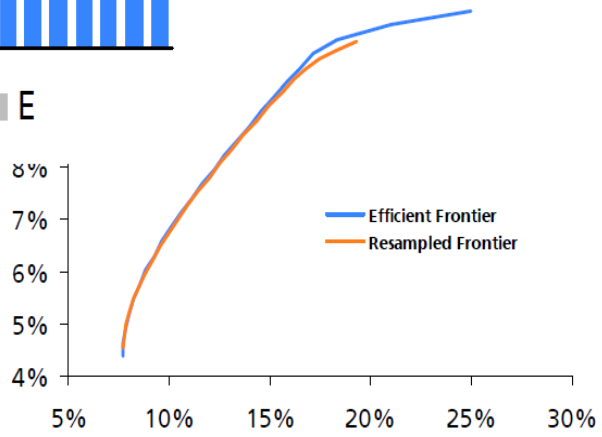
- Resampling is based on initial inputs $\mu 0$ and $V0$ which might be extreme
- Resampling only incorporates sample information
- Can lead to more allocation into risky assets

Resampled Portfolio - shortcomings

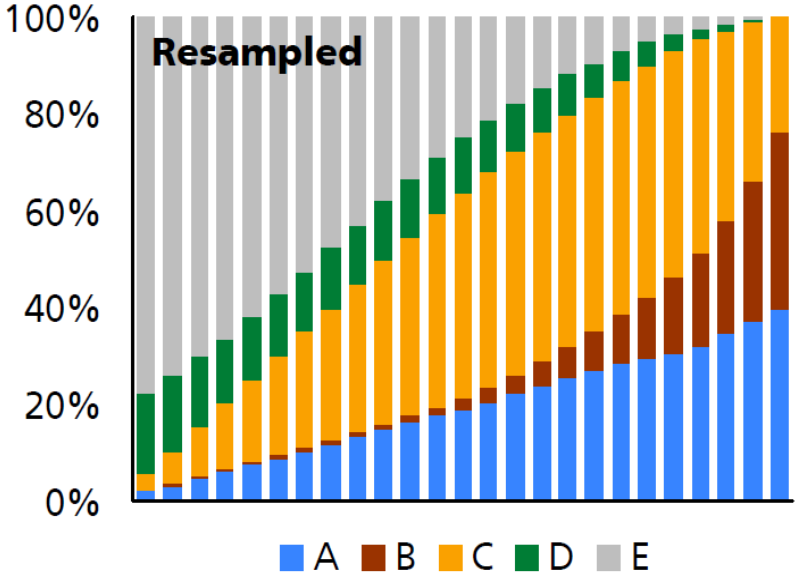
Volatility of asset B is 24%

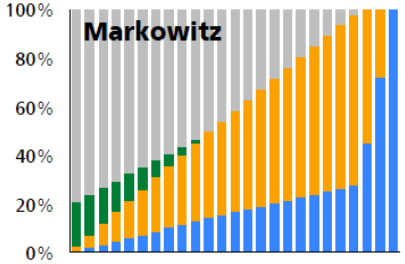


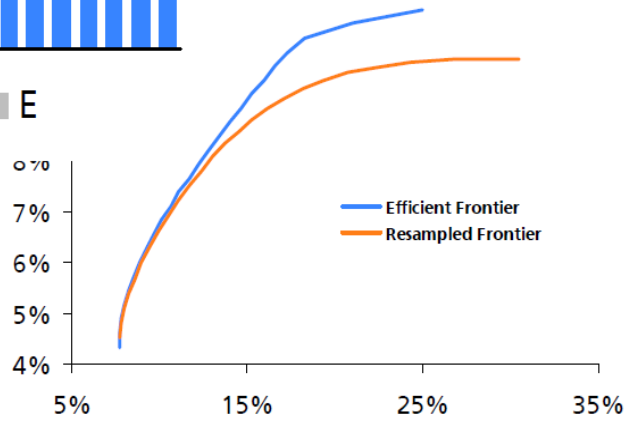




Volatility of asset B is 60%







Conclusion

- Classic mean variance optimisers tend to maximise the consequence of forecasting errors (Corner solution)
- Classic mean variance optimisers allocate portfolio weights without considering the statistical nature of the inputs
- The optimiser updates portfolio weights but does not investigate their statistical relevance
- The true inputs into an optimiser will never be known
- We have to use our experience and judgement
- There are tools that help to overcome the main shortcomings ...

Conclusion

- If we base the optimisation process only on information in the historic sample
  - Hard to overcome problem of parameter uncertainty
  - Never beat market consensus
- Irrational not to consider other information like
  - Experience
  - Superior knowledge due to market insights
  - Input from Investor, Statistician, Economist
- Main question is how to combine sample and non-sample data
  - How to update our prior believe with the information in the sample?
  - How to incorporate prior believes into MV optimiser?
  - How to create an intelligent optimising environment where new information is used to update our believes?

Another challenge for MVO

- Markowitz' mean-variance framework remains a mantra for quantitative portfolio construction. The process maximizes returns with respect to a given level of risk.

- A significant limitation with this framework is that the measure of risk is the standard deviation, which entails the implicit assumption that asset returns follow a multivariate normal distribution. However, it is known that the returns of financial assets do not follow a normal distribution. The volatility varies, or 'clusters' across time, resulting in non-stationary, heteroskedastic time-series. The high volatility clusters are associated with the fat tails.

Tail Risk Measure

- The primary challenges associated with this are threefold: having a reliable measure of tail risk, retaining a risk measure that is able to adapt to different market conditions, and a practical approach to building portfolios with consideration to these measures

Figure 2: Advantages and Disadvantages of VaR and CVaR as Measures of Tail Risk

Advantage	*Disadvantage*
◦ Simple and Intuitive ◦ VaR Measures often required for regulatory reporting ◦ CVaR is a 'coherent' risk measure	◦ VaR is not a 'coherent' risk measure ◦ VaR and CVaR don't account for co-dependency between assets

Source: Deutsche Bank Quantitative Strategy

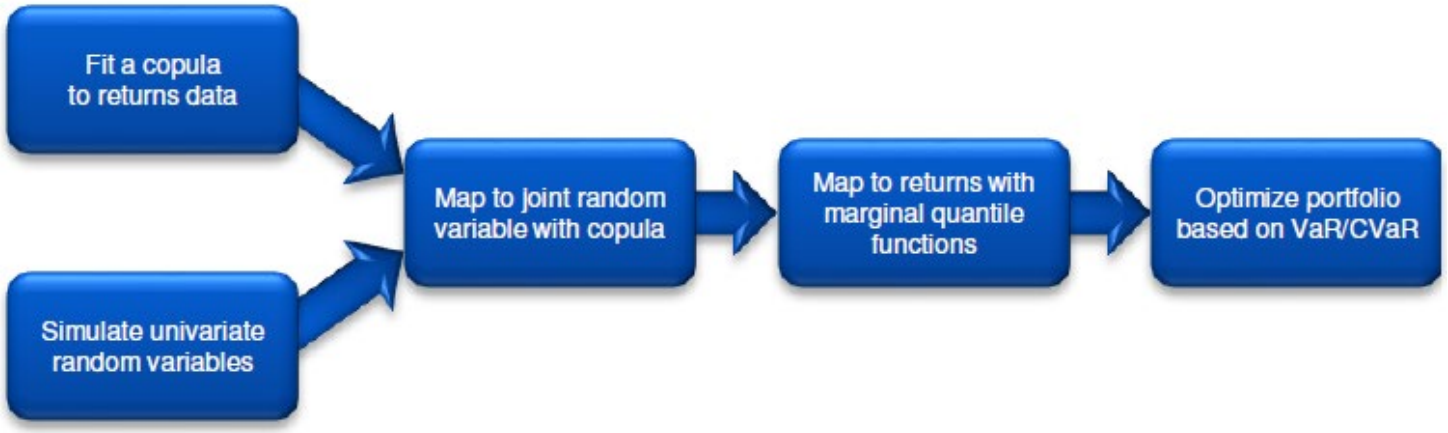
Figure 4: Advantages and Disadvantages of Sample Covariance Matrices as Measures of Tail Risk

Advantage	*Disadvantage*
◦ Often most accurate volatility forecasts ◦ Results are typically intuitive	◦ Does not specifically account for 'fat tails' ◦ Can be susceptible to outliers

Source: Deutsche Bank Quantitative Strategy

Mean-CVaR Optimization

- Copulas have received increased attention in quantitative finance in recent years. They are used to capture the co-dependence between marginal distributions of returns. Copulas are parametric, and each captures (tail) co-dependence in accordance with an underlying model. Once a given copula has been calibrated, returns are typically simulated via a Monte Carlo approach, and a measure such as VaR or CVaR is then optimized in determining a portfolio.



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